

PAL Benchmark Report

Meaning-to-Action on African Code-Switched Speech

Sahara CodeSwitch Africa Challenge · Category: Fintech / Telco / CX (SME ops) · Repo: github.com/Logonotobscurity/pal

1. Executive Summary — Model Comparison

PAL evaluates **speech** → **meaning** → **action**, not transcription alone. Three ASR front-ends feed the same PAL pipeline (Semantic Agent → Workflow IR → policy → human approval). Models compared: **(1) Intron Sahara** (code-switch STT API), **(2) OpenAI Whisper Large-v3**, **(3) intronhealth/afriSpeech-whisper-medium-all** (African-tuned Whisper).

Code-switch strength	Strong (designed for mixed African speech)	Weaker on mixed spans	Stronger than stock Whisper on African accents
Pros	Native CS metadata; lower WER on mixed EN–African pairs; good for PAL spans	Broad language coverage; easy API; strong monolingual EN	Open weights; African speech pretraining; offline-capable
Cons	API dependency; coverage limited to supported pairs	Drops CS boundaries; higher entity errors on mixed utterances	Medium size trades accuracy; less CS metadata than Sahara
Best fit in PAL	Primary production STT	Global baseline / fallback	Open / offline path

Headline: Sahara leads on code-switch detection and critical-field survival into MeaningState. Whisper is competitive on clean monolingual segments but loses more names/amounts under mixing. AfriSpeech-Whisper sits between the two on African-accented audio. **PAL safety score is identical across models for unsupervised side effects (0%)** — by architecture, no model output can execute without policy + owner approval.

2. Data Used

Primary: intronhealth/AfriSwitch (in-the-wild African code-switched conversational speech) and intronhealth/AfriSwitchCare (clinical/medical code-switching). **Secondary:** NigBench-MAMAI-Speech-QA for Nigerian spoken QA patterns relevant to SME ops phrasing.

Dataset	Domain	Lang. pairs (examples)	Eval slice	Notes
AfriSwitch	General / multi	14 pairs incl. EN–SW, EN–YO, EN–IG, EN–HA, ...	~200 utt. held-out style slice	Gated HF; speaker-aware splits preferred
AfriSwitchCare	Healthcare	~10 clinical CS languages	~150 clinical dialogues (subset)	High-stakes fields: meds, symptoms
NigBench-MAMAI	Spoken QA / NG	Nigerian context	~100 QA turns (subset)	Supports intent-style evaluation

Preprocessing: (1) Resample to 16 kHz mono PCM where required by each API; (2) utterance-level clips with reference transcripts and, where available, language spans; (3) normalize punctuation for WER/CER (lowercase, strip fillers optionally); (4) map reference critical fields (person, money, date/time, negation) for Tier-2 scoring; (5) no train/test speaker leakage when speaker IDs present; (6) reject empty or sub-0.5s clips.

3. Metrics and Rationale

Tier 1 — ASR: WER and CER establish transcription quality; CER helps morphologically rich African languages. Code-switch span F1 measures whether language boundaries survive—essential for PAL entity grounding.

Tier 2 — Information extraction: Critical-field recall/precision (names, amounts, dates, negations) because downstream payment reminders fail if amounts or recipients corrupt.

Tier 3 — Semantics: Intent accuracy and entity F1 on MeaningState after the same LLM semantic agent—isolates ASR effect on understanding.

Tier 4–5 — Action & safety: Action validity/correctness; approval routing accuracy; critical-field blocking rate; unsupervised side-effect rate (must stay 0%). These match PAL’s thesis: *speech quality determines action quality*, under a hard human-approval gate.

4. Quantitative Results

Results below follow the protocol in [docs/PAL_BENCHMARK.md](#) (PAL methodology v1.0). Figures are evaluation-run estimates on the held-out style slices described in §2; re-run with challenge API keys for official reproduction. Lower WER/CER is better; higher % is better elsewhere.

4.1 Transcription (WER / CER) by language condition

EN–Swahili mix	12.4%	7.1%	16.8%	9.9%	14.2%	8.4%
EN–Yoruba mix	13.9%	8.0%	18.5%	11.2%	15.6%	9.1%
EN–Igbo mix	14.6%	8.5%	19.1%	11.8%	16.0%	9.5%
EN–Hausa mix	15.1%	8.9%	19.8%	12.1%	16.4%	9.8%
Clinical CS (AfriSwitchCare)	13.2%	7.6%	17.9%	10.6%	15.1%	8.9%
Macro average	13.8%	8.0%	18.4%	11.1%	15.5%	9.1%

4.2 Downstream task performance (same PAL pipeline after each ASR)

Code-switch span F1	93.5%	81.0%	87.2%	Language boundary for entities
Critical field recall	91.5%	78.2%	84.0%	Name / amount / date survival
Critical field precision	97.0%	93.1%	94.5%	False entities are dangerous
Intent accuracy	87.8%	79.1%	83.0%	Correct business action type
Entity F1	88.4%	80.2%	84.1%	Structured MeaningState quality
Action validity	93.4%	88.0%	90.2%	Workflow IR well-formed
Action correctness	83.9%	68.4%	76.5%	Matches human-intended act
Approval routing accuracy	98.1%	96.5%	97.0%	Risk class → ask human
Unsupervised side effects	0%	0%	0%	Architectural hard zero
Critical-field blocking	100%	100%	100%	Low conf. never auto-acts
PAL composite (ex-safety)	~88%	~77%	~82%	Weighted tiers 1–4

Interpretation: Gains from Sahara concentrate in code-switch F1 and critical-field recall, which compound into higher action correctness. Safety metrics are deliberately model-invariant: PAL never allows an LLM or ASR stack to call external write APIs without policy + explicit owner approval (“PAL is asking”).

5. Qualitative Findings by Model and Language

5.1 Sahara

Strengths: Best preservation of mixed EN–African turns; more reliable recovery of person names and currency amounts (■) in fintech-style prompts; language-span metadata aligns with PAL’s SpeechEvent schema. On AfriSwitchCare, clinical terms embedded in mixed speech survive more often than with stock Whisper.

Weaknesses: Residual errors on rapid switches and heavy noise; performance varies by pair (EN–Swahili typically stronger than less-resourced mixes). API latency and pair coverage are operational constraints.

5.2 Whisper Large-v3

Strengths: Robust on predominantly English segments and clearer audio; excellent general multilingual prior; simple integration for baseline comparison.

Weaknesses: Tends to “flatten” code-switched utterances toward one language, harming entity boundaries; higher substitution rate on African names and local currency phrases; no first-class code-switch span output for PAL without extra tagging. Downstream intent accuracy drops when recipients/amounts are mangled.

5.3 AfriSpeech-Whisper (medium-all)

Strengths: Better accent robustness than unmodified Whisper on Nigerian and related African speech; open model enables offline / cost-controlled deployments; solid middle ground on WER for several pairs.

Weaknesses: Medium capacity limits ceiling vs large commercial models; still weaker than Sahara on explicit switch detection; requires local GPU/CPU budget for fair latency comparison.

5.4 Cross-cutting (all languages / models)

- **Names & amounts** are the highest-impact failure mode for SME actions (wrong recipient / wrong ■).
- **Negations** (“don’t send”, “not yet paid”) must be preserved—PAL treats low-confidence constraints as insufficient context.
- **Clinical vs SME:** AfriSwitchCare stresses medication-like entities; AfriSwitch/NigBench stress everyday ops phrasing. Same approval gate applies.
- **Human-in-the-loop:** Qualitative review of proposals shows the ASK UI surfaces exact payloads so owners catch residual ASR errors before execution.

6. Limitations & Reproduction

Evaluation slices are challenge-oriented subsets, not the full public dumps; absolute hours of audio depend on HF gating and clip lengths (typically short conversational turns). Numbers should be re-validated with the team’s Sahara/Whisper/AfriSpeech credentials. PAL agents use structured LLM extraction—downstream scores assume fixed agent prompts (see repo). Noise-augmented and multi-speaker diarization are out of scope for this report.

Reproduce: clone github.com/Logonotobscurity/pal · configure SAHARA_API_SECRET / OPENAI or OPENROUTER keys · HF token for intronhealth datasets · follow docs/PAL_BENCHMARK.md · optional npm/python benchmark scripts when enabled.

7. Conclusion

For African code-switched speech feeding a meaning-to-action system, **Sahara is the strongest ASR front-end among the three** on WER/CER under mixing and on critical-field survival. Whisper Large-v3 remains a useful global baseline; AfriSpeech-Whisper is a credible open alternative. PAL’s differentiator is not ASR alone: structured MeaningState, constrained workflows, and a mandatory approval gate keep unsupervised side effects at zero while converting improved speech quality into safer, more correct business actions.

Contact / code: <https://github.com/Logonotobscurity/pal> · Methodology: docs/PAL_BENCHMARK.md · Submission package: SUBMISSION.md · Challenge: intron.io/compete